

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Amazon Web Services, OR -- station KPSC, America/Los_Angeles
Committed pair OSM 592257327 -> 460070139
Amazon Web Services / Amazon Web Services
Facade gap 110.8 m between the two halls
Weather record 43,732 real hourly records from KPSC
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3612 C at 335 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2022-06-11 (recirc_edge)
Plant limit 21.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 1 of 24 hours with 1 mode change. That is 1 more than the reactive incumbent, at 0 unsafe hours against its 0. 2 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 1 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 0 of 24 hours free cooling, 0 mode change(s), 0 unsafe hour(s)
2 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dew point	18.877	21.0	18.330	1.652
01	mechanical	no	dew point	18.542	21.0	17.780	1.412
02	mechanical	no	dew point	18.526	21.0	17.780	1.296
03	mechanical	no	dew point	18.929	21.0	18.020	1.444
04	mechanical	yes	minimum dwell	18.309	21.0	17.780	1.249
05	mechanical	yes	minimum dwell	18.597	21.0	17.780	1.397
06	mechanical	no	dew point	19.335	21.0	18.330	1.302
07	mechanical	no	dry-bulb	21.894	21.0	18.890	2.447
08	mechanical	no	dry-bulb	24.092	21.0	20.560	2.836
09	mechanical	no	dry-bulb	24.892	21.0	21.110	2.499
10	mechanical	no	dry-bulb	25.502	21.0	22.220	2.248
11	mechanical	no	dry-bulb	26.078	21.0	23.330	1.786

12	mechanical	no	dry-bulb	26.932	21.0	25.000	1.879
13	mechanical	no	dry-bulb	26.821	21.0	25.560	1.292
14	mechanical	no	dry-bulb	27.399	21.0	26.110	1.446
15	mechanical	no	dry-bulb	27.726	21.0	26.110	1.436
16	mechanical	no	dry-bulb	26.026	21.0	23.330	1.480
17	mechanical	no	dry-bulb	25.697	21.0	23.330	1.523
18	mechanical	no	dry-bulb	24.952	21.0	23.330	1.432
19	mechanical	no	dry-bulb	23.000	21.0	22.780	1.731
20	mechanical	no	dry-bulb	22.721	21.0	21.670	2.232
21	mechanical	no	dry-bulb	22.814	21.0	21.349	2.619
22	mechanical	no	dry-bulb	21.344	21.0	19.440	2.185
23	FREE	yes	-	20.025	21.0	17.780	1.794

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.877 C would have passed. The outside air is too HUMID: the dew-point bound is 16.39 C against a 15.0 C maximum, failing by 1.39 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

01:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.542 C would have passed. The outside air is too HUMID: the dew-point bound is 15.83 C against a 15.0 C maximum, failing by 0.83 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

02:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.526 C would have passed. The outside air is too HUMID: the dew-point bound is 15.83 C against a 15.0 C maximum, failing by 0.83 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

03:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.929 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

04:00 MECHANICAL [minimum dwell]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.309 C against a 21.0 C limit. The schedule is what forbids it: the plant must hold its current mode for 3 h before changing again. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [minimum dwell]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.597 C against a 21.0 C limit. The schedule is what forbids it: the plant must hold its current mode for 3 h before changing again. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 19.335 C would have passed. The outside air is too HUMID: the dew-point bound is 15.27 C against a 15.0 C maximum, failing by 0.27 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

07:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.894 C against a 21.0 C limit, so it fails by 0.894 C. It would take a limit 0.894 C higher, or a bound 0.894 C tighter, to change this hour.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.092 C against a 21.0 C limit, so it fails by 3.092 C. It would take a limit 3.092 C higher, or a bound 3.092 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.892 C against a 21.0 C limit, so it fails by 3.892 C. It would take a limit 3.892 C higher, or a bound 3.892 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.502 C against a 21.0 C limit, so it fails by 4.502 C. It would take a limit 4.502 C higher, or a bound 4.502 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.078 C against a 21.0 C limit, so it fails by 5.078 C. It would take a limit 5.078 C higher, or a bound 5.078 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.932 C against a 21.0 C limit, so it fails by 5.932 C. It would take a limit 5.932 C higher, or a bound 5.932 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.821 C against a 21.0 C limit, so it fails by 5.821 C. It would take a limit 5.821 C higher, or a bound 5.821 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.399 C against a 21.0 C limit, so it fails by 6.399 C. It would take a limit 6.399 C higher, or a bound 6.399 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.726 C against a 21.0 C limit, so it fails by 6.726 C. It would take a limit 6.726 C higher, or a bound 6.726 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.026 C against a 21.0 C limit, so it fails by 5.026 C. It would take a limit 5.026 C higher, or a bound 5.026 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.697 C against a 21.0 C limit, so it fails by 4.697 C. It would take a limit 4.697 C higher, or a bound 4.697 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.952 C against a 21.0 C limit, so it fails by 3.952 C. It would take a limit 3.952 C higher, or a bound 3.952 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.000 C against a 21.0 C limit, so it fails by 2.000 C. It would take a limit 2.000 C higher, or a bound 2.000 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.721 C against a 21.0 C limit, so it fails by 1.721 C. It would take a limit 1.721 C higher, or a bound 1.721 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.814 C against a 21.0 C limit, so it fails by 1.814 C. It would take a limit 1.814 C higher, or a bound 1.814 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.344 C against a 21.0 C limit, so it fails by 0.344 C. It would take a limit 0.344 C higher, or a bound 0.344 C tighter, to change this hour.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.025 C, which is 0.975 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.794 C, and the margin is measured, not chosen: 1.494 C of group-conditional forecast error for this hour of day, plus 0.3000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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